

Optimal Convergence of the Fixed-Point Method for Nonlinear Eddy Current Problems

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A method is presented for the optimal choice of the fixed-point reluctivity in nonlinear eddy current problems solved for steady-state in the frequency and the time domain. The optimal selection is shown to ensure a minimal contraction number for the fixed-point method. The proposed technique is compared to earlier approaches using a 2-D example.

Index Terms—Convergence of numerical methods, eddy currents, Fourier transforms, numerical analysis.

I. INTRODUCTION

A METHOD yielding an improved convergence of the fixed-point method when applied to nonlinear eddy current problems in the frequency domain has been introduced in [1]. The basic idea is to change the fixed-point reluctivity, ν_{FP} , at each iteration step in each integration point. The suggestion in [1] is to set ν_{FP} to the maximal value of the reluctivity over a time period, T , at the previous iteration step.

The speed of the convergence of the fixed-point method increases if the contraction number of the iteration is smaller [2]. An analysis of the dependence of the contraction number on the fixed-point reluctivity is carried out in this paper. It is found that it is possible to choose a value for ν_{FP} resulting in a smaller contraction number than that yielded by the approach in [1].

Furthermore, a method is presented to get the steady state solution in the time domain using fixed-point techniques and the fixed-point reluctivity resulting in optimal convergence is found for this case, too.

The convergence rates obtained by the new methods are compared to those of [1] using a 2-D numerical example.

II. SPECIFICATION OF THE PROBLEM

Maxwell's equations in the quasi-static case are

$$\text{curl} \mathbf{H} = \mathbf{J} \quad (1)$$

$$\text{curl} \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (2)$$

$$\text{div} \mathbf{B} = 0 \quad (3)$$

where $\mathbf{H}$ is the magnetic field intensity, $\mathbf{J}$ is the current density, $\mathbf{B}$ is the flux intensity, $\mathbf{E}$ is the electric field intensity.

The relevant material relationships are

$$\mathbf{H} = \nu \mathbf{B}, \quad \mathbf{J} = \sigma \mathbf{E} + \mathbf{J}_0 \quad (4)$$

where $\mathbf{J}_0$ is the impressed current density, ν is the reluctivity and σ is the conductivity.

Introducing the magnetic vector potential $\mathbf{A}$ and the modified electric scalar potential v , which is used to ensure the symmetry of the system matrix

$$\mathbf{B} = \text{curl} \mathbf{A}, \quad \mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t} - \text{grad} \frac{\partial v}{\partial t}, \quad (5)$$

and using the law of continuity, the following partial differential equations are obtained:

$$\begin{aligned} \text{curl}(\nu \text{curl} \mathbf{A}) + \sigma \left(\frac{\partial \mathbf{A}}{\partial t} + \text{grad} \frac{\partial v}{\partial t} \right) &= \mathbf{J}_0, \\ \text{div} \left(\sigma \frac{\partial \mathbf{A}}{\partial t} + \sigma \text{grad} \frac{\partial v}{\partial t} \right) &= 0. \end{aligned} \quad (6)$$

Applying Galerkin's method to (6) results in an ordinary differential equation system

$$\mathbf{M}(\sigma) \dot{x}(t) + \mathbf{S}(\nu) x(t) = f(t) \quad (7)$$

where $\mathbf{M}(\sigma)$ is the mass-matrix, $\mathbf{S}(\nu)$ is the stiffness-matrix, the vector $x(t)$ represents the unknown time dependent potentials $\mathbf{A}$ and v , and $f(t)$ contains the excitation. The stiffness matrix depends on ν linearly, but ν is a function of x . Since $\mathbf{M}(\sigma)$ depends linearly on σ and σ is independent of x , the simple notation $\mathbf{M}$ will be used.

The excitation $f(t)$ is assumed to be a periodic function, i.e., $f(t) = f(t+T)$ with $T > 0$ being the time period. The steady state solution we are looking for also satisfies the periodicity condition $x(t) = x(t+T)$.

III. THE FIXED-POINT METHOD

The fixed-point method is a technique for solving nonlinear equations such as (7). A fixed-point parameter is introduced to linearize the equations with the nonlinearity brought to the right-hand-side. Since, in (7), the nonlinearity is due to the dependence of ν on the solution x , it is obvious to define a fixed-point reluctivity ν_{FP} as $\nu = \nu_{\text{FP}} + (\nu - \nu_{\text{FP}})$. Substituting this into the first equation in (4), (6) becomes

$$\begin{aligned} \text{curl}(\nu_{\text{FP}} \text{curl} \mathbf{A}) + \sigma \left(\frac{\partial \mathbf{A}}{\partial t} + \text{grad} \frac{\partial v}{\partial t} \right) \\ = \mathbf{J}_0 + \text{curl}[(\nu_{\text{FP}} - \nu) \text{curl} \mathbf{A}], \\ \text{div} \left(\sigma \frac{\partial \mathbf{A}}{\partial t} + \sigma \text{grad} \frac{\partial v}{\partial t} \right) = 0. \end{aligned} \quad (8)$$

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Galerkin's method applied to (8) leads to the ordinary differential equation system

$$\mathbf{M}\dot{x}(t) + \mathbf{S}(\nu_{\text{FP}})x(t) = f(t) + \mathbf{S}(\nu_{\text{FP}} - \nu)x(t). \quad (9)$$

Equation (9) implies an iterative method, namely

$$\mathbf{M} \frac{dx^{(s+1)}}{dt} + \mathbf{S}(\nu_{\text{FP}}^{(s)})x^{(s+1)} = f + \mathbf{S}(\nu_{\text{FP}}^{(s)} - \nu^{(s)})x^{(s)}. \quad (10)$$

where s is the iteration count, $\nu^{(s)}$ is determined by $x^{(s)}$ and $\nu_{\text{FP}}^{(s)}$ will be obtained from $\nu^{(s)}$ to speed up the convergence of (10).

In each iteration step, a linear ordinary differential equation system has to be solved. Two different methods to get the solution of (10) are discussed below.

A. Method in the Frequency Domain

The basic idea of the harmonic balance method [3] is that if the trigonometric Fourier coefficients of two functions are equal, then, under appropriate conditions, these two functions are equal.

To solve (10) in each fixed-point iteration step, the following two tasks have to be carried out:

1) Solve the equation system

$$[j\omega k \mathbf{M} + \mathbf{S}(\nu_{\text{FP}})]X_k^{(s+1)} = \mathbf{F}_k(f + \mathbf{S}(\nu_{\text{FP}} - \nu)x^{(s)}) \quad (11)$$

where $\mathbf{F}_k(x)$ denotes the k th trigonometric Fourier coefficient

$$\mathbf{F}_k(x) = \hat{x}_k = \frac{1}{T} \int_0^T x e^{-j\omega k t} dt. \quad (12)$$

2) Calculate the solution of the $(s+1)$ th iteration step from the Fourier coefficients:

$$x^{(s+1)} = \mathbf{F}^{-1}(X_k^{(s+1)}) \quad (13)$$

where

$$\mathbf{F}^{-1}(X_k) = \sum_{k=-\infty}^{\infty} X_k e^{j\omega k t}. \quad (14)$$

The function x can depend not only on time, but on the space coordinates too, therefore X_k will depend on the space coordinates for each k .

B. Method in the Time Domain

A very similar method to that of [4] is presented here except that, instead of the weighted Euler scheme used there, a central difference scheme is employed because of its higher order accuracy.

The discretization of (10) with the central difference scheme using equidistant time steps within one period ($\Delta t = T/n$) leads to n coupled simultaneous linear equation systems

$$\mathbf{M} \frac{x_{k+1}^{(s+1)} - x_{k-1}^{(s+1)}}{2\Delta t} + \mathbf{S}(\nu_{\text{FP}}^{(s)})x_k^{(s+1)} = f_k + \mathbf{S}(\nu_{\text{FP}}^{(s)} - \nu_k^{(s)})x_k^{(s)} \quad (15)$$

where $x_k = x(k\Delta t)$ and $f_k = f(k\Delta t)$, $k = 0, 1, \dots, n-1$.

Equation (15) results in a huge block-structured equation system with a block-circulant matrix [5] and a block-structured right hand-side which can be solved by a block-diagonalization method. This leads to a result similar to that obtained by the harmonic balance method:

1) Solve the equation

$$\left[j \sin\left(\frac{i}{n} 2\pi\right) \frac{\mathbf{M}}{\Delta t} + \mathbf{S}(\nu_{\text{FP}}) \right] Y_i = \mathbf{D}\mathbf{F}_i(f_k + \mathbf{S}(\nu_{\text{FP}} - \nu_k)x_k^{(r)}) \quad (16)$$

where $\mathbf{D}\mathbf{F}_i$ ($i = 0, 1, \dots, n-1$) is the i th term in the discrete Fourier transform, namely

$$\mathbf{D}\mathbf{F}_i(x_k) = \frac{1}{n} \sum_{k=0}^{n-1} x_k e^{-j2\pi \cdot i \frac{k}{n}}. \quad (17)$$

2) Transform back the solutions in the k th time step into the time domain

$$x_k^{(r+1)} = \mathbf{D}\mathbf{F}_k^{-1}(Y_i) \quad (18)$$

where $\mathbf{D}\mathbf{F}_k^{-1}$ is the symbol for the inverse discrete Fourier transformation:

$$\mathbf{D}\mathbf{F}_k^{-1}(Y_i) = \sum_{i=0}^{n-1} Y_i e^{j2\pi \cdot k \frac{i}{n}}. \quad (19)$$

In this case x_k and Y_i can depend on the space coordinates for all k and i .

IV. THE FIXED-POINT THEOREM

The convergence of the iterative procedure (10) can be proven by the fixed-point theorem. This states that a sequence generated by a function $a_{n+1} = g(a_n)$ is convergent for any initial element a_0 if g is a contraction i.e., there exists a contraction number $q \in (-1, 1)$ so that for any x and y

$$\|g(x) - g(y)\| \leq q \|x - y\| \quad (20)$$

where $\|\cdot\|$ is a suitable norm. Furthermore, the limit value of the sequence is independent of a_0 .

In the harmonic balance case, (13) and (11) describe the generator function as

$$g(x) = \mathbf{F}^{-1}[j\omega k \mathbf{M} + \mathbf{S}(\nu_{\text{FP}})]^{-1} \mathbf{F}_k(f + \mathbf{S}(\nu_{\text{FP}} - \nu)x). \quad (21)$$

Similarly, (18) and (16) yield the generator function for the time domain method. Since the variable x depends both on time and space, two norms are necessary. They are the L_2 norms

$$\|x\|_{L_2(\Omega)} = \sqrt{\int_{\Omega} x^2 d\Omega}, \|x\|_{L_2(0,T)} = \sqrt{\int_0^T x^2 dt}. \quad (22)$$

V. ANALYSIS OF THE CONTRACTION NUMBER

The generator functions of the harmonic balance and the time domain method are very similar, so the former one will be investigated only. The results obtained are directly applicable to the time domain case, too.

In order to get the contraction number, the norm of the difference of two generator functions has to be estimated (see (20)). Equation (21) yields

$$g(x) - g(y) = \mathbf{F}^{-1}[\mathbf{j}\omega k \mathbf{M} + \mathbf{S}(\nu_{\text{FP}})]^{-1} \mathbf{F}_k \times (f + \mathbf{S}(\nu_{\text{FP}} - \nu)(x - y)). \quad (23)$$

For the estimation, it is necessary to investigate the norm of the Fourier transform (12) and of its inverse (14).

It can be easily shown by using the orthogonality of the trigonometric functions that (14) has the property

$$\|\mathbf{F}^{-1} \hat{f}_k\|_{L_2(\Omega)} \|L_2(0,T)\| = \|\mathbf{F}^{-1} \hat{f}_k\|_{L_2(\Omega)} \|L_2(0,T)\| \quad (24)$$

where $\hat{f}_k$ denotes a sequence of Fourier coefficients (see (12)).

The property of (13) we need is

$$(\mathbf{F}_k x) \cdot (\mathbf{F}_k y) = \mathbf{F}_k(x * y), \quad (25)$$

where $x * y$ is the convolution of two functions, i.e.,

$$(x * y)(t) = \int_0^T x(s)y(t-s)ds. \quad (26)$$

The product of two sequences on the left-hand-side of (25) is a sequence whose elements are the products of the elements of the original sequences, i.e., $(a_k) \cdot (b_k) = (a_k b_k)$.

The last used property is the convolution inequality [6]

$$\|x * y\|_{L_p(0,T)} \leq \|x\|_{L_p(0,T)} \|y\|_{L_1(0,T)} \quad (27)$$

where

$$\|x\|_{L_p(0,T)} = \begin{cases} \sqrt[p]{\frac{1}{T} \int_0^T |x|^p dt}, & 1 \leq p < \infty \\ \sup\{|x(t)| : 0 \leq t < T\}, & p = \infty. \end{cases} \quad (28)$$

Using the properties (24), (25) and (27) as well as the Cauchy inequality [7], the estimation of (23) can be carried out

$$\begin{aligned} & \|g(x) - g(y)\|_{L_2(\Omega)} \|L_2(0,T)\| \\ & \leq C \left\| \frac{\nu_{\text{FP}} - \nu}{\nu_{\text{FP}}} \right\|_{L_2(0,T)} \|x - y\|_{L_2(\Omega)} \|L_2(0,T)\| \end{aligned} \quad (29)$$

$$\leq C \left\| \frac{\nu_{\text{FP}} - \nu}{\nu_{\text{FP}}} \right\|_{L_\infty(0,T)} \|x - y\|_{L_2(\Omega)} \|L_2(0,T)\|. \quad (30)$$

It follows from (29) and (30) that the contraction number can be estimated as

$$q \leq C \left\| \frac{\nu_{\text{FP}} - \nu}{\nu_{\text{FP}}} \right\|_{L_2(0,T)} \quad (31)$$

$$\text{or} \\ q \leq C \left\| \frac{\nu_{\text{FP}} - \nu}{\nu_{\text{FP}}} \right\|_{L_\infty(0,T)}. \quad (32)$$

where C is a constant, coming from the mathematical deduction of (29). It is near to 1 and independent of ν_{FP} .

The estimations (31) and (32) indicate that the contraction number strongly depends on the fixed-point reluctivity. The optimal value can be determined by minimizing the right-hand side of (31) or (32). Both cases will be investigated.

A. The Minimization of the Contraction Number Using the L_2 Norm

To minimize (31) with the L_2 norm, the derivative of the right-hand-side of (31) with respect to ν_{FP} , must be set to zero. The result is

$$\nu_{\text{FP}} = \int_0^T \nu^2(t) dt / \int_0^T \nu(t) dt. \quad (33)$$

B. The Minimization of the Contraction Number Using the L_∞ Norm

The right-hand-side of (32) is not differentiable if we use the infinite norm (L_∞), but it can easily be shown that its minimal value is

$$\nu_{\text{FP}} = \frac{\min_{t \in [0,T]}(\nu) + \max_{t \in [0,T]}(\nu)}{2}. \quad (34)$$

VI. ANALYSIS OF THE CONVERGENCE

At the estimation of the contraction number, we have not yet taken into account that the reluctivity is a function of the solution. As shown in [8], a stable method of updating the reluctivity leads to a fixed-point iteration, too. The change of the reluctivity in the s th iteration step can be expressed as

$$\nu^{(s+1)} - \nu^{(s)} = (q - 1) \left(1 - B'(H)\nu^{(s)} \right) \left(\nu^{(s)} - \nu^{(s-1)} \right) \quad (35)$$

where q is the contraction number established by (31) and (32), B' is the derivative of the B - H curve, H is an approximate value of the magnetic field intensity and $\nu^{(s)}$ is the reluctivity, at the s th iteration step.

Equation (35) yields a condition for the fixed-point reluctivity

$$\|\nu(1 - B'(H)\nu)\|_{L_\infty(0,T)} < \nu_{\text{FP}}. \quad (36)$$

VII. THE OPTIMAL FIXED-POINT RELUCTIVITY

The optimal fixed-point reluctivity can be determined from the conditions (33), (34) and (36). Our numerical investigations show that the best choice is

$$\nu_{\text{FP}}^{\text{opt}} := \max \left\{ \frac{\int_0^T \nu^2 dt}{\int_0^T \nu dt}, \frac{\min_{t \in [0,T]}(\nu) + \max_{t \in [0,T]}(\nu)}{2} \right\}. \quad (37)$$

However, if this value does not fulfil the condition (36), it has to be changed to

$$\nu_{\text{FP}}^{\text{opt}} := \|\nu(1 - B'(H)\nu)\|_{L_\infty(0,T)}. \quad (38)$$

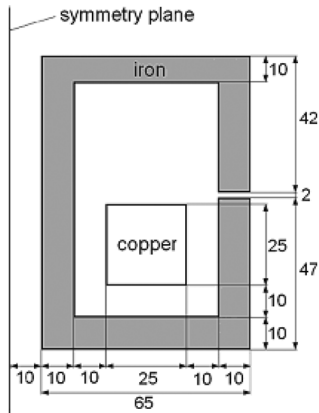


Fig. 1. The geometry of the 2-D problem, which is used to compare the different methods. The lengths are given in millimeter.

The analysis of the convergence carried out in Sections V and VI shows that larger values than the one indicated in (36), (38) ensure convergence, but lead to a higher number of nonlinear iterations. Selecting a smaller value for the fixed-point parameter, the method will not be convergent and results in oscillations during the nonlinear iteration steps.

The condition given in (36) for ν_{FP} depends strongly on the B - H curve. For some strongly saturating curves it can lead to ν_{FP} values very near to the maximal value of ν proposed in [1]. In the problem treated in Section VIII, this condition leads to a value of 0.75–0.8 times the maximum of ν .

VIII. NUMERICAL EXAMPLE

The 2-D numerical example consists of a copper conductor within a conducting ferromagnetic screen (see Fig. 1.). The copper conductor ($\sigma = 5.7 \cdot 10^7$ S/m) is driven by a sinusoidal voltage of 50 Hz. The conductivity of the ferromagnetic screen is 10^6 S/m. The B - H curve is specified in [4].

To compare the methods, the saturation has been increased by using different values for the exciting voltage per unit length (see Table I.). The numbers of the necessary nonlinear iterations are shown in Table I, using 9 harmonics for the harmonic balance method and 20 time steps per period for the method in the time domain. The initial value of ν_{FP} has been chosen as $1/(2 \mu_0)$ where μ_0 is the permeability of vacuum. If the peak value of the excitation is low, there is little difference between the methods, but at higher excitations, the saturation increases, and the speed increment of the fixed-point method is appreciable.

IX. CONCLUSION

On summary, the fixed-point method results in a stable iterative procedure for solving the eddy-current problem with nonlinear material properties.

It has been shown how to choose the optimal fixed-point reluctivity to accelerate the convergence. The numerical example

TABLE I
NUMBER OF FIXED-POINT ITERATIONS

Method	Voltage per unit length			
	1.4 V/m	3 V/m	5 V/m	7 V/m
$\nu_{FP}^{\max}$				
Harmonic Balance	54	71	110	117
ν_{FP}^{opt}				
Harmonic Balance	54	67	106	108
$\nu_{FP}^{\max}$				
Time Domain	48	62	107	167
ν_{FP}^{opt}				
Time Domain	48	57	100	144

Number of fixed-point iterations with different voltages per unit length, using both the harmonic balance method and the time domain method presented. $\nu_{FP}^{\max}$ is the choice of [1], ν_{FP}^{opt} is the choice of the optimal fixed-point reluctivity. The convergence criterion is the same as in [1].

indicates that the speed of the convergence can be over 10% higher if the optimal fixed-point parameter is used instead of the choice of [1], because the contraction number is smaller.

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